

Notes: Gandhi, Navarro, and Rivers (JPE, 2020)

On the Identification of Gross Output Production Functions

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1 Big picture

- **Question.** Can gross output production functions be identified under the standard proxy-variable / dynamic-panel logic when intermediate inputs are flexible, and if not, what extra structure is needed?
- **Setting.** A short panel of firms with gross output Y , capital K , labor L , and intermediate inputs M , where some inputs are flexible and others are predetermined or dynamic.
- **Core challenge.** In gross output, intermediate inputs enter both the production function and the proxy equation. With only output and input quantities, the usual proxy-variable inversion does not supply enough independent variation to separate productivity from the flexible-input term.
- **Main negative result.** Standard proxy-variable and dynamic-panel strategies generally *do not* nonparametrically identify gross output production functions unless there is additional exogenous variation in flexible-input demand, such as prices.
- **Main positive result.** The paper develops a new nonparametric identification strategy that uses the firm's first-order condition for flexible inputs. A nonparametric share regression identifies the flexible-input elasticity, and then a partial-differential-equation argument plus lagged-input moments identify the rest of the production function.
- **Empirical takeaway.** In Chilean and Colombian plant data, the method yields sensible input elasticities and much more plausible productivity dispersion than OLS. OLS systematically overstates the intermediate-input elasticity and understates capital/labor elasticities and productivity heterogeneity.
- **Why this paper matters.** It gives a clean identification foundation for gross output estimation, which is crucial in applications where intermediate inputs matter directly: importing, markups, misallocation, and production networks.

2 Data and measurement (key objects)

2.1 Observed firm-level panel

- **Unit of observation.** Firm j in period t .
- **Observed variables.** Gross output Y_{jt} , capital K_{jt} , labor L_{jt} , and intermediate inputs M_{jt} ; lowercase letters denote logs $(y_{jt}, k_{jt}, l_{jt}, m_{jt})$.
- **Asymptotics.** Short panel asymptotics: number of firms $J \rightarrow \infty$ with fixed T .
- **Information structure.** Inputs may be predetermined, dynamic, or flexible. The paper's main setup treats intermediate inputs as the flexible input and capital/labor as nonflexible inputs chosen before the within-period ex post shock.

2.2 Unobservables and timing

- **Productivity decomposition.** Total productivity is

$$\nu_{jt} = q_{jt} + \varepsilon_{jt},$$

where q_{jt} is known to the firm when choosing inputs and ε_{jt} is an ex post shock observed only after choices are made.

- **Persistent productivity.** q_{jt} follows a Markov process,

$$\Pr(q_{jt} \mid I_{j,t-1}) = \Pr(q_{jt} \mid q_{j,t-1}), \quad h(q_{j,t-1}) \equiv \mathbb{E}[q_{jt} \mid q_{j,t-1}].$$

- **Flexible-input demand.** Intermediate inputs are chosen using period- t information, so they can respond directly to q_{jt} .

2.3 Empirical applications

- **Colombia.** Manufacturing census, all plants with more than 10 employees, 1981–1991.
- **Chile.** Manufacturing census, all firms with more than 10 employees, 1979–1996.
- **Industries.** The paper estimates separate production functions for the five largest three-digit industries: food products (311), textiles (321), apparel (322), wood products (331), and fabricated metal products (381), plus an aggregate manufacturing specification.
- **Variable construction.** Real gross output is deflated revenue; intermediates combine raw materials, energy, and services; labor is a wage-weighted sum of blue- and white-collar workers; capital is built by perpetual inventory.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production function and productivity

The production technology is

$$Y_{jt} = F(K_{jt}, L_{jt}, M_{jt})e^{\nu_{jt}} \iff y_{jt} = f(k_{jt}, l_{jt}, m_{jt}) + \nu_{jt}, \quad (1)$$

with $\nu_{jt} = q_{jt} + \varepsilon_{jt}$.

- **Interpretation.** This is a gross output production function, so intermediates enter directly as an input, unlike value-added specifications.
- **Key modeling feature.** The ex ante component q_{jt} creates transmission bias because firms choose inputs knowing q_{jt} , while the econometrician does not observe it.
- **Ex post shock.** ε_{jt} is orthogonal to the information set used to choose inputs and is the source of identifying variation in the first-order-condition strategy.

3.2 Flexible input demand and the proxy-variable setup

The proxy-variable literature assumes the flexible input demand takes the form

$$m_{jt} = M_t(k_{jt}, l_{jt}, q_{jt}), \quad (2)$$

strictly monotone in the single unobservable q_{jt} .

Under perfect competition in output and intermediate-input markets, the firm's problem for M_{jt} is

$$\max_{M_{jt}} P_t \mathbb{E} \left[F(k_{jt}, l_{jt}, m_{jt}) e^{q_{jt} + \varepsilon_{jt}} \mid I_{jt} \right] - r_t M_{jt}, \quad (3)$$

so the first-order condition is

$$P_t \frac{\partial}{\partial M_{jt}} F(k_{jt}, l_{jt}, m_{jt}) e^{q_{jt}} E = r_t, \quad (4)$$

where $E \equiv \mathbb{E}[e^{\varepsilon_{jt}}]$. Hence the flexible-input demand can be written as

$$m_{jt} = M(k_{jt}, l_{jt}, q_{jt} - d_t) = M_t(k_{jt}, l_{jt}, q_{jt}), \quad d_t \equiv \ln(r_t/P_t) - \ln E. \quad (5)$$

- **Interpretation.** The usual LP/ACF logic says: invert intermediate-input demand to recover q_{jt} and then use lagged inputs as instruments in a second stage.
- **Why gross output is hard.** Since m_{jt} is itself in the production function, the proxy equation and production function are tightly linked. The same source of variation that helps proxy productivity is also absorbed by the flexible-input term.

3.3 Why the standard proxy-variable / dynamic-panel approach fails

The proxy-variable first stage identifies only the random object

$$\bar{f}(k_{jt}, l_{jt}, m_{jt}) \equiv f(k_{jt}, l_{jt}, m_{jt}) + M^{-1}(k_{jt}, l_{jt}, m_{jt}) + d_t, \quad (6)$$

not the structural elasticities separately. In the second stage,

$$y_{jt} = f(k_{jt}, l_{jt}, m_{jt}) + h(f(k_{j,t-1}, l_{j,t-1}, m_{j,t-1}) + d_{t-1} - f(k_{j,t-1}, l_{j,t-1}, m_{j,t-1})) + \eta_{jt} + \varepsilon_{jt}, \quad (7)$$

with lagged inputs serving as instruments.

The identifying restriction can be written as

$$\mathbb{E}[y_{jt} \mid G_{jt}] = \mathbb{E}[f(k_{jt}, l_{jt}, m_{jt}) \mid G_{jt}] + h(f(k_{j,t-1}, l_{j,t-1}, m_{j,t-1}) + d_{t-1} - f(k_{j,t-1}, l_{j,t-1}, m_{j,t-1})), \quad (8)$$

where the instrument set includes current predetermined inputs and lagged output and inputs,

$$G_{jt} = (k_{jt}, l_{jt}, d_{t-1}, y_{j,t-1}, k_{j,t-1}, l_{j,t-1}, m_{j,t-1}, \dots).$$

- **Theorem 1.** If relative prices are constant over time ($d_t = d$ for all t), the model is not identified. There is a continuum of observationally equivalent alternatives:

$$\tilde{f} = (1-a)f^0 + af, \quad \tilde{h}(x) = ad + (1-a)h^0\left(\frac{x-ad}{1-a}\right), \quad a \in (0, 1).$$

- **Economic intuition.** Once current and lagged nonflexible inputs are conditioned on, the only remaining source of variation in m_{jt} is the productivity innovation η_{jt} and aggregate price variation d_t . If d_t is constant, nothing is left that can separately identify the flexible-input term.
- **Corollary 1.** Even if prices vary over time, with only $T = 2$ periods the model is still nonidentified.
- **Corollary 2.** If either the production function or the productivity law of motion varies over time, aggregate price variation can no longer save identification.

3.4 Main positive result: identify the flexible-input elasticity from revenue shares

The key step is to transform the flexible-input first-order condition into a share equation. Taking logs of the first-order condition and subtracting the production function gives

$$s_{jt} = \ln E + \ln \left(\frac{\partial f(k_{jt}, l_{jt}, m_{jt})}{\partial m_{jt}} \right) - \varepsilon_{jt} = \ln D^E(k_{jt}, l_{jt}, m_{jt}) - \varepsilon_{jt}, \quad (9)$$

where

$$s_{jt} \equiv \ln \left(\frac{r_t M_{jt}}{P_t Y_{jt}} \right)$$

is the log intermediate-input revenue share.

Hence the conditional expectation of the share regression is

$$\mathbb{E}[s_{jt} \mid k_{jt}, l_{jt}, m_{jt}] = \ln D^E(k_{jt}, l_{jt}, m_{jt}). \quad (10)$$

Then

$$E = \mathbb{E} \left[\exp(\ln D^E(k_{jt}, l_{jt}, m_{jt}) - s_{jt}) \right], \quad (11)$$

and the flexible-input elasticity is identified as

$$\frac{\partial f(k_{jt}, l_{jt}, m_{jt})}{\partial m_{jt}} = \frac{D^E(k_{jt}, l_{jt}, m_{jt})}{E}. \quad (12)$$

- **Theorem 2.** Under assumptions 1–4 and observed relative prices r_t/P_t , the nonparametric share regression identifies the flexible-input elasticity almost everywhere.
- **Why this is powerful.** The usual proxy approach cannot isolate the flexible-input elasticity. The share equation does exactly that, because the ex post shock ε_{jt} drops out in conditional expectation.
- **Economic intuition.** Revenue shares reveal marginal products. In Cobb-Douglas this is familiar; the paper shows the idea survives fully nonparametrically.

3.5 Recovering the full production function from a partial differential equation

Once the flexible-input elasticity is known, integrate it in m :

$$\int \frac{\partial f(k_{jt}, l_{jt}, m_{jt})}{\partial m_{jt}} dm_{jt} = f(k_{jt}, l_{jt}, m_{jt}) + C(k_{jt}, l_{jt}). \quad (13)$$

Define the observable transformed output

$$Y_{jt}^* \equiv y_{jt} - \varepsilon_{jt} - \int \frac{\partial f(k_{jt}, l_{jt}, m_{jt})}{\partial m_{jt}} dm_{jt} = -C(k_{jt}, l_{jt}) + q_{jt}. \quad (14)$$

Using the Markov property of q_{jt} ,

$$Y_{jt}^* = -C(k_{jt}, l_{jt}) + h(Y_{j,t-1}^* + C(k_{j,t-1}, l_{j,t-1})) + \eta_{jt}, \quad (15)$$

so

$$\mathbb{E}[Y_{jt}^* \mid k_{jt}, l_{jt}, Y_{j,t-1}^*, k_{j,t-1}, l_{j,t-1}] = -C(k_{jt}, l_{jt}) + h(Y_{j,t-1}^* + C(k_{j,t-1}, l_{j,t-1})). \quad (16)$$

- **Assumption 5.** There must be enough support variation in current predetermined inputs (k_{jt}, l_{jt}) conditional on lagged transformed output and lagged inputs, so current and lagged pieces can be separated.
- **Theorem 3.** If the flexible-input elasticity is known and the support condition holds, the production function is nonparametrically identified up to an additive constant.
- **Interpretation.** The share regression gives the m -dimension of the production function; lagged-input moments identify the remaining integration constant in (k, l) and the productivity law of motion.

3.6 Estimator implementation

The paper proposes a two-step sieve estimator.

- **Step 1: share regression.** Estimate $D^E(k, l, m)$ with a complete polynomial series of degree r in (k, l, m) , recover $\hat{\varepsilon}_{jt}$ and the flexible-input elasticity.
- **Step 2: integrate and estimate (C, h) .** Integrate the estimated elasticity analytically, construct $\hat{Y}_{jt}^*$, and estimate the integration constant $C(k, l)$ and Markov process $h(\cdot)$ with polynomial sieves.
- **Baseline implementation.** In the Monte Carlo and main applications, the paper uses degree-2 polynomials for the elasticity and $C(\cdot)$, and a degree-3 polynomial for $h(\cdot)$.
- **Inference.** The estimator is computationally simple; for functionals of interest, standard two-step asymptotics and a nonparametric bootstrap are used.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Core failure.** In gross output, the proxy equation and production function are too intertwined. Without additional exogenous demand shifters for the flexible input, the standard second stage cannot distinguish productivity from the flexible-input contribution.
- **Aggregate prices are not a practical solution.** In theory, time-series variation in relative prices can help. In practice, the paper's Monte Carlo shows identification is weak unless price variation is implausibly large and panels are very long.
- **Dynamic panel has the same fundamental problem.** Under the same basic structure, dynamic-panel estimators also lack enough cross-sectional variation to identify gross output nonparametrically. Their advantage is only that they do not require scalar unobservability.

4.2 What identifies the model in the new approach

- **First-order-condition content.** The share regression exploits the economic restriction implied by optimal flexible-input choice, not just timing restrictions.

- **Ex post shock orthogonality.** Because ε_{jt} is realized after the input choice, it is mean independent of current inputs in the relevant sense. This is what turns the share regression into an identifying equation.
- **Support/rank condition.** After integrating the flexible-input elasticity, enough variation in current predetermined inputs relative to lagged states is needed to separate $C(k, l)$ from $h(\cdot)$.
- **No need for observed firm-specific prices.** Unlike price-based approaches, the new strategy works even when only quantity/expenditure data are observed.

4.3 Relation to the literature

- **Against LP/ACF in gross output.** The paper shows exactly why the gross-output extension of Levinsohn–Petrin / Akerberg–Caves–Frazer is not identified without extra price-type variation.
- **Relative to Doraszelski–Jaumandreu.** Those papers use observed price variation plus parametric cross-equation restrictions. Gandhi–Navarro–Rivers recover the same broad economic link nonparametrically and without needing firm-level prices.
- **Relative to Griliches–Ringstad.** The paper is a nonparametric generalization of the old insight that cost/revenue shares reveal flexible-input elasticities, combined with modern dynamic-panel / proxy timing restrictions.

5 Tables and figures

5.1 Figure 1: Monte Carlo on proxy-variable identification using only time-series price variation

Read:

- The figure studies a proxy-variable estimator applied to gross output under different amounts of time-series variation in relative intermediate-input prices and different panel lengths.
- Even with 20–50 periods, realistic aggregate price variation is not enough to recover the true intermediate-input elasticity reliably.
- Only when price variation is increased to roughly 10 times the baseline seen in the data do the estimates begin to converge toward the truth.
- This figure is the practical version of Theorem 1: aggregate price variation can identify in theory, but is often too weak in practice.

5.2 Table 1: Monte Carlo performance of the proposed estimator

Read:

- The table compares true elasticities with estimates from the paper’s nonparametric procedure under three DGPs: Cobb–Douglas, CES, and translog.
- Under Cobb–Douglas, the estimator recovers mean intermediate and capital elasticities almost exactly (roughly 0.650 and 0.250) with essentially zero mass outside the $(0, 1)$ range.

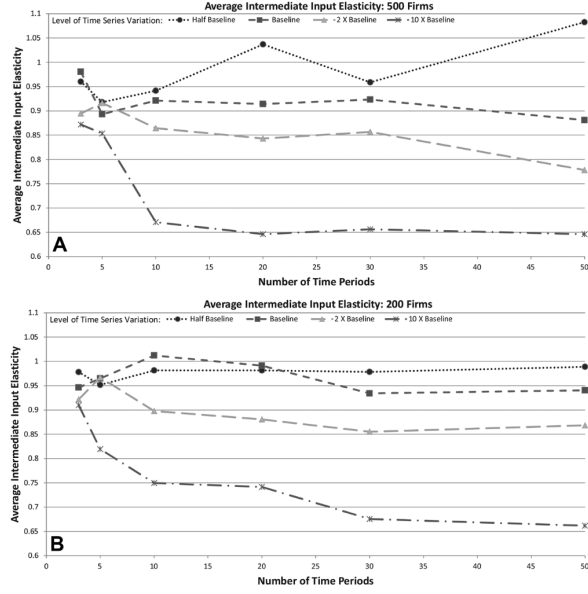


FIG. 1.—Monte Carlo-proxy variable estimator applied to gross output. This figure presents the results from applying a proxy variable estimator extended to gross output to Monte Carlo data generated as described in section III and appendix O4. The data are generated under four different levels of time-series variation. The X-axis measures the number of time periods in the panel used to generate the data. The Y-axis measures the average of the estimated intermediate-input elasticity across 100 Monte Carlo simulations. The true value of the elasticity is 0.65. Panel A includes 500 firms; panel B includes 200 firms. A color version of this figure is available online.

TABLE 1
MONTE CARLO—GNR ESTIMATOR PERFORMANCE

	TRUE FUNCTIONAL FORM					
	Cobb-Douglas		CES		Translog	
	At True Parameters (1)	GNR Estimates (2)	At True Parameters (3)	GNR Estimates (4)	At True Parameters (5)	GNR Estimates (6)
Intermediates:						
Average mean elasticity	.6500	.6502 (.0015)	.6747 (.0027)	.6746 (.0030)	.6574 (.0007)	.6572 (.0014)
Average standard deviation	0	.0038 (.0012)	.1197 (.0014)	.1193 (.0018)	.0321 (.0004)	.0324 (.0012)
Average fraction outside of (0,1)	0	.0000 (.0000)	0	.0000 (.0000)	.0000 (.0000)	.0000 (.0000)
Capital:						
Average mean elasticity	.2500	.2504 (.0065)	.2253 (.0027)	.2198 (.0071)	.2263 (.0006)	.2263 (.0082)
Average standard deviation	0	.0091 (.0043)	.1197 (.0014)	.1210 (.0022)	.0333 (.0004)	.0349 (.0022)
Average fraction outside of (0,1)	0	.0000 (.0000)	0	.0096 (.0055)	.0000 (.0000)	.0000 (.0000)
Sum:						
Average mean elasticity	.9000	.9006 (.0065)	.9000	.8943 (.0074)	.8836 (.0002)	.8835 (.0080)
Average standard deviation	0	.0093 (.0044)	0	.0227 (.0037)	.0056 (.0001)	.0108 (.0062)

NOTE.—In this table, we compare estimates of the production function elasticities using our nonparametric procedure with estimates using the true values. We simulate data from three different parametric production functions: Cobb-Douglas, CES, and translog. See

- Under CES and translog, it also matches the mean and distribution of elasticities closely, so the procedure works both when elasticities are constant and when they are heterogeneous.
- The estimator remains accurate even though it is nonparametric and the data-generating processes differ materially.

5.3 Appendix Monte Carlo: robustness to adjustment costs

Read:

- The paper also introduces adjustment costs for the flexible input in simulation, which violates the exact static first-order-condition assumption.
- Dynamic-panel estimators improve as adjustment costs get larger because lagged intermediates then create useful variation.
- The paper's estimator still performs remarkably well even with large adjustment costs, showing that the first-order-condition approach is empirically robust to moderate misspecification.

5.4 Table 2: Chilean and Colombian production elasticities

Read:

- Across the five main 3-digit industries and the aggregate specification, the proposed estimator produces sensible elasticities: intermediates are largest (roughly 0.50–0.67), labor next (roughly 0.22–0.52), and capital smallest (roughly 0.04–0.16).

- The model’s additional moment restrictions are mostly not rejected in industry-by-industry tests, except some evidence of misspecification for fabricated metals in Chile.
- Allowing for fixed effects leaves intermediate-input elasticities unchanged by construction and produces broadly similar results, though precision can decline in smaller Colombian samples.
- Allowing additional unobservables in flexible-input demand or relaxing the independence assumption for the ex post shock leaves the core results largely unchanged.

6 Short summary

- **Conclusion.** Standard gross-output proxy-variable and dynamic-panel estimators are generically not identified without extra flexible-input-demand shifters.
- **Identification insight.** Revenue shares plus the flexible-input first-order condition nonparametrically identify the flexible-input elasticity; integrating that elasticity and using lagged-input moments identifies the rest of the production function.
- **Empirical lesson.** In real plant data, correcting the gross-output identification problem materially changes estimated elasticities and raises estimated productivity dispersion relative to OLS.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Gross output production functions are not automatically identified by the standard proxy-variable logic. Flexible intermediate inputs create a nonidentification problem unless there is extra exogenous variation in their demand.
- **Methodological contribution.** The paper’s new estimator combines an index-number style share argument with a dynamic-panel style lagged-state argument. This produces a practical two-step nonparametric estimator for gross output.
- **Quantitative implication.** The difference between the new method and OLS is large in practice. Average output elasticities under OLS are biased by at least 23% and in some cases by as much as 73%, while OLS also understates productivity heterogeneity.
- **Empirical credibility.** The method performs well in Monte Carlo simulations under Cobb–Douglas, CES, and translog technologies, and remains robust when the exact static first-order-condition assumption is perturbed by adjustment costs.

7.2 Economic intuition

- Flexible inputs react immediately to expected productivity. That is why they are useful as proxies, but also why they cause trouble in gross output: the same source of variation drives both the proxy and the production function.

- Revenue shares provide the missing link because they reveal the marginal product of the flexible input. Once that margin is pinned down, the remaining part of the production function can be backed out from the dynamic evolution of productivity and predetermined inputs.
- The paper therefore turns what was previously a nuisance — the firm’s static input-choice equation — into the central identifying restriction.

7.3 Takeaway

This paper is best read as a foundation paper for gross output estimation. Its main message is not only that standard methods fail, but also exactly *why* they fail and how to fix them. The fix is: use the economics of optimal flexible-input choice to recover the flexible-input elasticity directly, then use the panel structure to recover everything else. For anyone using gross output to study import competition, markups, networks, or misallocation, this is the conceptual benchmark.